

● MEDIUM

● PROBLEM-SOLVING AND DATA ANALYSIS

● ANSWER KEY

Two-variable data: Models and scatterplots

01

10 answers · Rationale-rich · Teach from doc

Q1

B

B. -0.8

Choice B is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. Thus, the slope of the line of best fit shown is negative. The slope of a line passing through two points, x_1, y_1 and x_2, y_2 , can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points 1, 12 and 11, 4. Substituting 1, 12 and 11, 4 for x_1, y_1 and x_2, y_2 , respectively, in $\frac{y_2 - y_1}{x_2 - x_1}$ gives $\frac{4 - 12}{11 - 1}$, which is equivalent to $-\frac{8}{10}$, or -0.8. Therefore, of the given choices, -0.8 is closest to the slope of the line of best fit shown.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Q2

A

A. 1

Choice A is correct. The data point with $x = 1$ has a y -coordinate of 12. The y -value predicted by the line of best fit at $x = 1$ is approximately 11. The difference between the y -coordinate of the data point and the y -value predicted by the line of best fit at $x = 1$ is $12 - 11$, or 1.

Choices B and C are incorrect and may result from incorrectly reading the scatterplot. Choice D is incorrect. This is the y -coordinate of the data point at $x = 1$.

Q3**B****B. -1.1**

Choice B is correct. A line in the xy -plane that passes through points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points 0, 14 and 13, 0. It follows that the slope of this line of best fit is approximately $\frac{0 - 14}{13 - 0}$, or $-\frac{14}{13}$. Of the given choices, -1.1 is closest to $-\frac{14}{13}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q4**C****C. -0.7**

Choice C is correct. A line of best fit is shown in the scatterplot such that as the value of x increases, the value of y decreases. It follows that the slope of the line of best fit shown is negative. The slope of a line in the xy -plane that passes through the points x_1, y_1 and x_2, y_2 can be calculated as $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points 0, 8 and 10, 1. Substituting 0, 8 for x_1, y_1 and 10, 1 for x_2, y_2 in $\frac{y_2 - y_1}{x_2 - x_1}$ yields the slope of the line being approximately $\frac{1 - 8}{10 - 0}$, which is equivalent to $-\frac{7}{10}$, or -0.7. Therefore, of the given choices, -0.7 is the closest to the slope of this line of best fit.

Choice A is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice B is incorrect. The line of best fit shown has a negative slope, not a positive slope.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**A****A.** 0.60

Choice A is correct. A line in the xy -plane that passes through the points (x_1, y_1) and (x_2, y_2) has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes through the points with approximate coordinates $(0, 3.2)$ and $(2, 4.4)$. It follows that the slope of the line of best fit is approximately $\frac{4.4 - 3.2}{2 - 0}$, which is equivalent to $\frac{1.2}{2}$, or 0.6. Therefore, of the given choices, 0.6 is closest to the slope of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A****A.** $n = 16,090 + 4,680t$

Choice A is correct. An equation of a line of best fit can be written in the form $y = a + bx$, where a is the y -intercept of the line and b is the slope. In the scatterplot shown, the line of best fit intersects the y -axis just over halfway between 10,000 and 20,000, or approximately 16,000. The line of best fit also intersects the graph at $(5, 40,000)$. Using the slope formula

$b = \frac{y_2 - y_1}{x_2 - x_1}$ and two points that lie on the graph such as $(5, 40,000)$ and $(0, 16,000)$, the slope

can be approximated as $\frac{40,000 - 16,000}{5 - 0}$, or 4,800. Only choice A has a y -intercept near the

estimate of 16,000 and a slope near the estimate of 4,800. Therefore, an equation of the line of best fit could be $n = 16,090 + 4,680t$.

Choice B is incorrect because the values for the slope and the y -coordinate of the y -intercept are switched. Choice C is incorrect because the value for the slope is approximately double the actual slope. Choice D is incorrect because the values for the slope and the y -intercept are switched and because the slope is approximately double the actual slope.

Q7**C**

C. $y = 3.30x + 0.82$

Choice C is correct. A line that most closely fits the data is a line with an approximately balanced number of data points above and below the line. Fitting a line to the data shown results in a line with an approximate slope of 3 and a y-intercept near the point $(0, 1)$. An equation for the line can be written in slope-intercept form, $y = mx + b$, where m is the slope and b is the y-coordinate of the y-intercept. The equation $y = 3.30x + 0.82$ in choice C fits the data most closely.

Choices A and B are incorrect because the slope of the lines of these equations is 0.82, which is a value that is too small to be the slope of the line that fits the data shown. Choice D is incorrect. The graph of this equation has a y-intercept at $(0, -3.30)$, not $(0, 0.82)$. This line would lie below all of the data points, and therefore would not closely fit the data.

Q8**C**

C. 1.8

Choice C is correct. A line in the xy -plane that passes through points x_1, y_1 and x_2, y_2 has a slope of $\frac{y_2 - y_1}{x_2 - x_1}$. The line of best fit shown passes approximately through the points $(0, 0.2)$ and $(5, 9.3)$. It follows that the slope of this line is approximately $\frac{9.3 - 0.2}{5 - 0}$, which is equivalent to $\frac{9.1}{5}$, or 1.82. Therefore, of the given choices, 1.8 is closest to the slope of the line of best fit shown.

Choice A is incorrect. This value is closest to the y-coordinate of the y-intercept of the line of best fit shown.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**D**

D. $y = 9.4 - 0.9x$

Choice D is correct. The data points suggest that as the variable x increases, the variable y decreases, which implies that an appropriate linear model for the data has a negative slope. The data points also show that when x is close to 0, y is greater than 9. Therefore, the y -intercept of the graph of an appropriate linear model has a y -coordinate greater than 9. The graph of an equation of the form $y = a + bx$, where a and b are constants, has a y -intercept with a y -coordinate of a and has a slope of b . Of the given choices, only choice D represents a graph that has a negative slope, -0.9 , and a y -intercept with a y -coordinate greater than 9, 9.4 .

Choice A is incorrect. The graph of this equation has a positive slope, not a negative slope, and a y -intercept with a y -coordinate less than 1, not greater than 9.

Choice B is incorrect. The graph of this equation has a y -intercept with a y -coordinate less than 1, not greater than 9.

Choice C is incorrect. The graph of this equation has a positive slope, not a negative slope.

Q10**C**

C. Increasing exponential

Choice C is correct. Because the value of the investment increases each year, the function that best models how the value of the investment changes over time is an increasing function. It's given that each year, the value of the investment increases by 0.49% of its value the previous year. Since the value of the investment changes by a fixed percentage each year, the function that best models how the value of the investment changes over time is an exponential function. Therefore, the function that best models how the value of the investment changes over time is an increasing exponential function.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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